

# IIR Design, Bilinear Transformation, & Data Preprocessing

From analog prototypes to stable digital filters—and the phase choices hidden in ML preprocessing



## Where this 75-minute lecture is going

### 0–15 min • Analog prototypes

Butterworth, Chebyshev I/II, and elliptic tradeoffs.

### 15–40 min • Bilinear transform

Map the analog LHP into the digital unit disk while preserving stability.

### 40–60 min • Warping + prewarp

Derive  $\Omega(\omega)$  and hit the desired digital critical frequencies.

### 60–75 min • ML preprocessing

Causal IIR versus forward-backward zero-phase filtering and deployment implications.

**DSP bridge: IIR design gives efficient recursive filters; ML preprocessing must also account for phase, causality, and train/deployment consistency.**

Classical DSP supplies structure and constraints; machine learning either reuses that structure or learns parameters inside it.

## Why accept feedback and nonlinear phase?

### Efficiency

Sharp magnitude selectivity can often be obtained with far fewer coefficients than an FIR meeting comparable magnitude specifications.

### Tradeoff

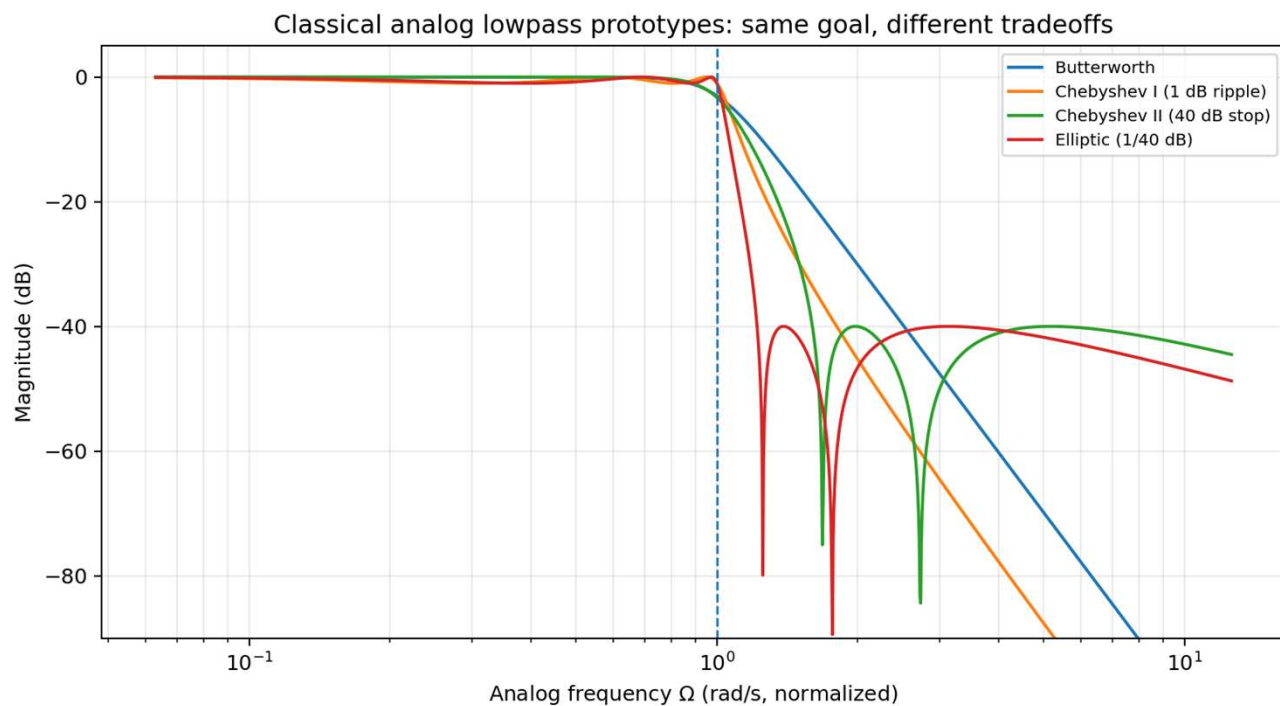
The recursive denominator introduces poles and typically nonlinear phase / frequency-dependent group delay.

### ML preprocessing

For offline denoising or smoothing, compact IIR filters are attractive—but preprocessing must match the temporal assumptions of deployment.

$$H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}}$$

## Four classical ways to spend ripple and transition width



### Butterworth

Maximally flat magnitude in the passband; monotonic.

### Chebyshev

Spend ripple in one band to sharpen the transition.

### Elliptic

Ripple in both bands; steepest transition for a given order/specification.

## Maximally flat magnitude response

$$|H_c(j\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}}$$

$$|H_c(j\Omega_c)| = \frac{1}{\sqrt{2}} \Rightarrow -3.01 \text{ dB}$$

### Order N

Higher N steepens the transition around  $\Omega_c$ .

### No passband ripple

Butterworth is attractive when smooth magnitude behavior matters more than the narrowest possible transition.

### Good answer

At the cutoff, the squared magnitude is 1/2 regardless of N.

## What increasing order buys—and what it costs

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$$\Omega \gg \Omega_c : |H_c(j\Omega)| \approx \left( \frac{\Omega_c}{\Omega} \right)^N$$

### Magnitude slope

Each additional pole increases asymptotic attenuation by about 20 dB/decade.

### Implementation cost

Higher order means more states / sections and greater numerical sensitivity in direct-form implementations.

### Practice

Use second-order sections for moderate/high-order digital IIR implementations when possible.

## Where the ripple is allowed to live

| Prototype    | Passband   | Stopband   | Transition / comment                     |
|--------------|------------|------------|------------------------------------------|
| Butterworth  | monotonic  | monotonic  | maximally flat; smooth                   |
| Chebyshev I  | equiripple | monotonic  | sharper for allowed passband ripple      |
| Chebyshev II | monotonic  | equiripple | spend ripple in stopband                 |
| Elliptic     | equiripple | equiripple | steepest transition for given order/spec |

### Engineering habit

Choose the prototype from passband/stopband tolerances and order requirements—not from a universal ranking.

## CONCEPT CHECK

# Which prototype sounds plausible?

### A • Sensor calibration

Passband amplitude must be smooth;  
moderate transition width is acceptable.

### B • Tight transition

A small amount of controlled passband  
ripple is acceptable to lower order.

### C • Most aggressive

Ripple is acceptable in both bands and  
minimum order is important.

### Good answer

Butterworth is a natural first choice  
because it is monotonic and maximally flat.

### Good answer

Chebyshev I trades passband ripple for a  
sharper transition.

### Good answer

Elliptic is often the most aggressive  
classical choice.



## What must a useful domain map preserve?

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### Stability

Analog stable poles satisfy  $\text{Re}\{s\} < 0$ . They must map to  $|z| < 1$ .

### Frequency axis

The analog  $j\Omega$  axis should map to the digital unit circle  $z = e^{j\omega}$ .

### One-to-one mapping

Avoid aliasing of multiple analog frequencies onto the same digital frequency.

**The bilinear transform satisfies these properties—but its price is nonlinear frequency warping.**

## The substitution that maps the half-plane to the disk

$$s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}$$

$$z = \frac{1 + sT/2}{1 - sT/2}$$

### Interpret T

T is the sampling period used in the analog-to-digital mapping.

### Important

This is not the impulse-invariance map  $z = e^{sT}$ . The bilinear transform avoids frequency aliasing but warps the axis.

## Why the analog left-half plane lands inside the unit circle

$$|z|^2 = \frac{|1 + sT/2|^2}{|1 - sT/2|^2}$$

$$s = \sigma + j\Omega, \sigma < 0 \Rightarrow |1 + sT/2| < |1 -$$

$$\therefore |z| < 1$$

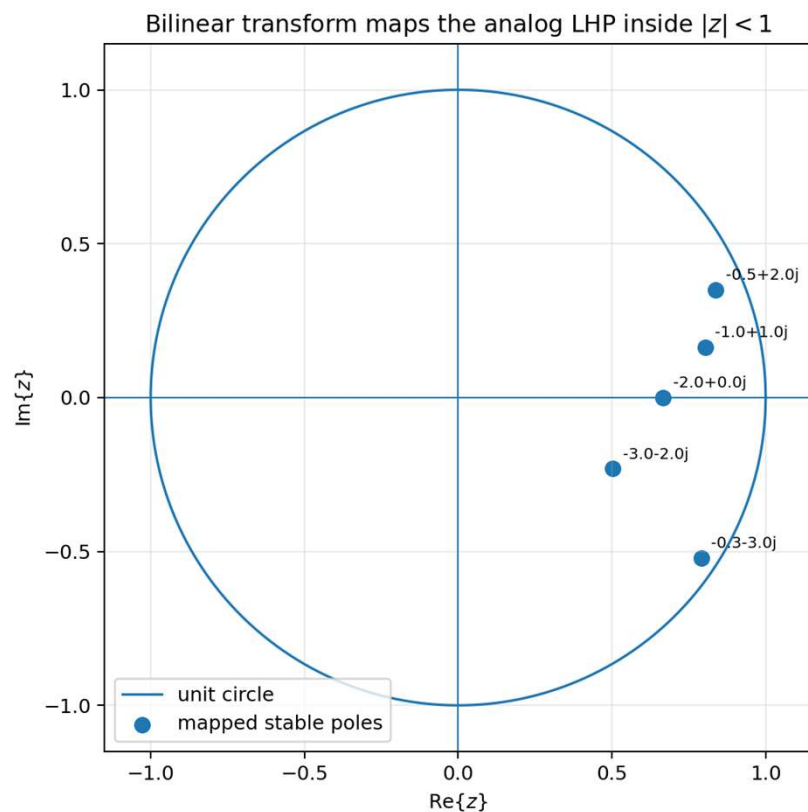
### Good answer

The negative real part makes the numerator magnitude smaller than the denominator magnitude, so a stable analog pole maps inside the digital unit circle.

### Boundary

For  $\sigma=0$ , numerator and denominator magnitudes are equal, hence  $|z|=1$ .

## Representative analog stable poles map inside $|z|=1$



### Interpretation

Every shown analog pole has negative real part. After the bilinear map, every digital pole lies strictly inside the unit circle.

### This is the guarantee we want

Start from a stable analog prototype and the mapped digital filter remains stable, barring numerical implementation issues.

## The analog $j\Omega$ axis maps exactly onto the unit circle

$$s = j\Omega, \quad z = e^{j\omega}$$

$$j\Omega = \frac{2}{T} \frac{1 - e^{-j\omega}}{1 + e^{-j\omega}}$$

$$\frac{1 - e^{-j\omega}}{1 + e^{-j\omega}} = j \tan\left(\frac{\omega}{2}\right)$$

### Endpoints

$\Omega=0$  maps to  $\omega=0$ . As  $|\Omega| \rightarrow \infty$ , the digital frequency approaches  $\pm\pi$  and  $z$  approaches  $-1$ .

### No aliasing

The entire analog frequency axis maps one-to-one into  $-\pi < \omega < \pi$ .

## The price of the one-to-one map

$$\Omega = \frac{2}{T} \tan \left( \frac{\omega}{2} \right)$$

$$\omega = 2 \tan^{-1} \left( \frac{\Omega T}{2} \right)$$

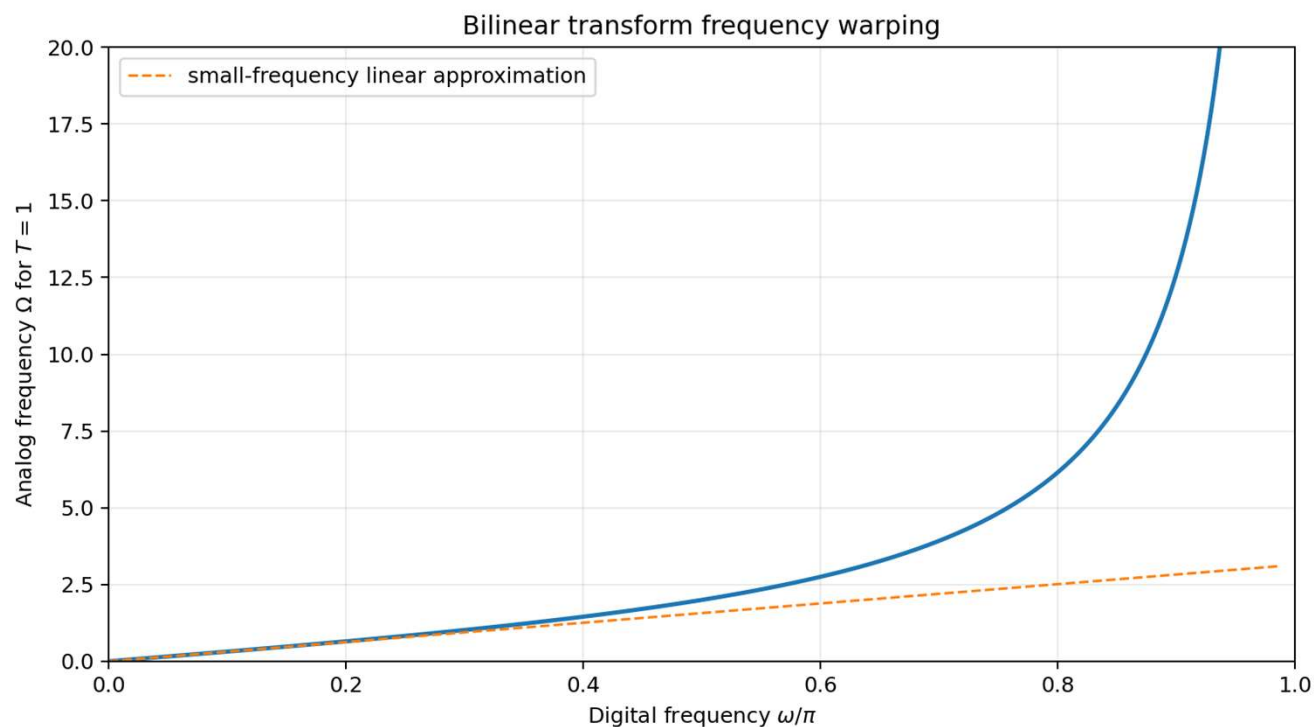
### Low frequency

For small  $\omega$ ,  $\tan(\omega/2) \approx \omega/2$ , so  $\Omega \approx \omega/T$ .

### Near Nyquist

The map becomes strongly nonlinear; equal analog frequency intervals do not remain equal digital intervals.

## Near Nyquist, the analog axis is stretched dramatically



### Read the curve

The relationship is nearly linear near DC but diverges as  $\omega \rightarrow \pi$ .

### Design consequence

If you design the analog cutoff at the unwrapped numeric value, the final digital cutoff misses its target.

## PREWARPING

# Design the analog prototype at the frequency that maps to the desired digital edge

$$\Omega_k = \frac{2}{T} \tan \left( \frac{\omega_k}{2} \right)$$

## Design sequence

### 1 • Digital spec

$\omega_p$ ,  $\omega_s$ , ripple / attenuation

### 2 • Prewarp

map each critical digital frequency to  $\Omega$

### 3 • Analog design

design Butterworth / Chebyshev / elliptic prototype

### 4 • Bilinear map

substitute  $s(z)$  to obtain the digital filter



### WORKED EXAMPLE

## Prewarp a desired cutoff at $0.4\pi$ rad/sample

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$$\omega_c = 0.4\pi, \quad T = 1$$

$$\Omega_c = 2 \tan(0.2\pi) \approx 1.4531 \text{ rad/s}$$

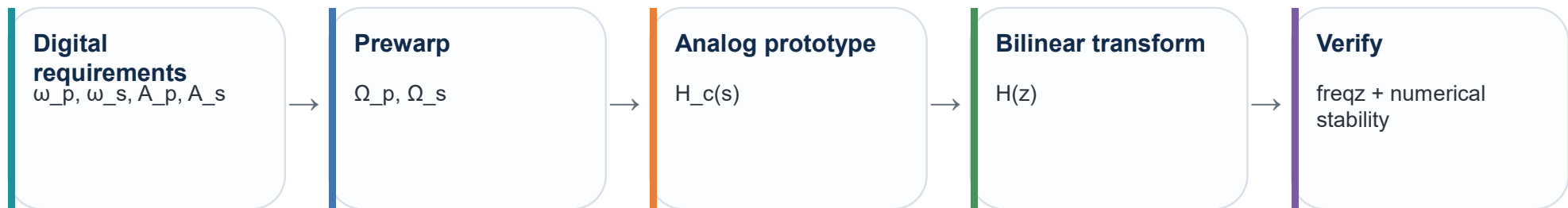
### Meaning

Design the analog prototype at  $\Omega_c \approx 1.4531$ —not at  $0.4\pi \approx 1.2566$ .

### Check

After the bilinear transform, the measured  $-3$  dB point lands at approximately  $0.4\pi$ .

## From digital specification to digital IIR



**The bilinear transform preserves stability, not exact frequency spacing. Verification is part of the design, not an optional afterthought.**

## Explicit analog prototype → bilinear transform in SciPy

```
wc = 0.4*np.pi
T = 1.0
Omega_c = (2/T)*np.tan(wc/2)

b_a, a_a = signal.butter(4, Omega_c, analog=True)
b_z, a_z = signal.bilinear(b_a, a_a, fs=1/T)
w, H = signal.freqz(b_z, a_z)
```

### Why show the explicit path?

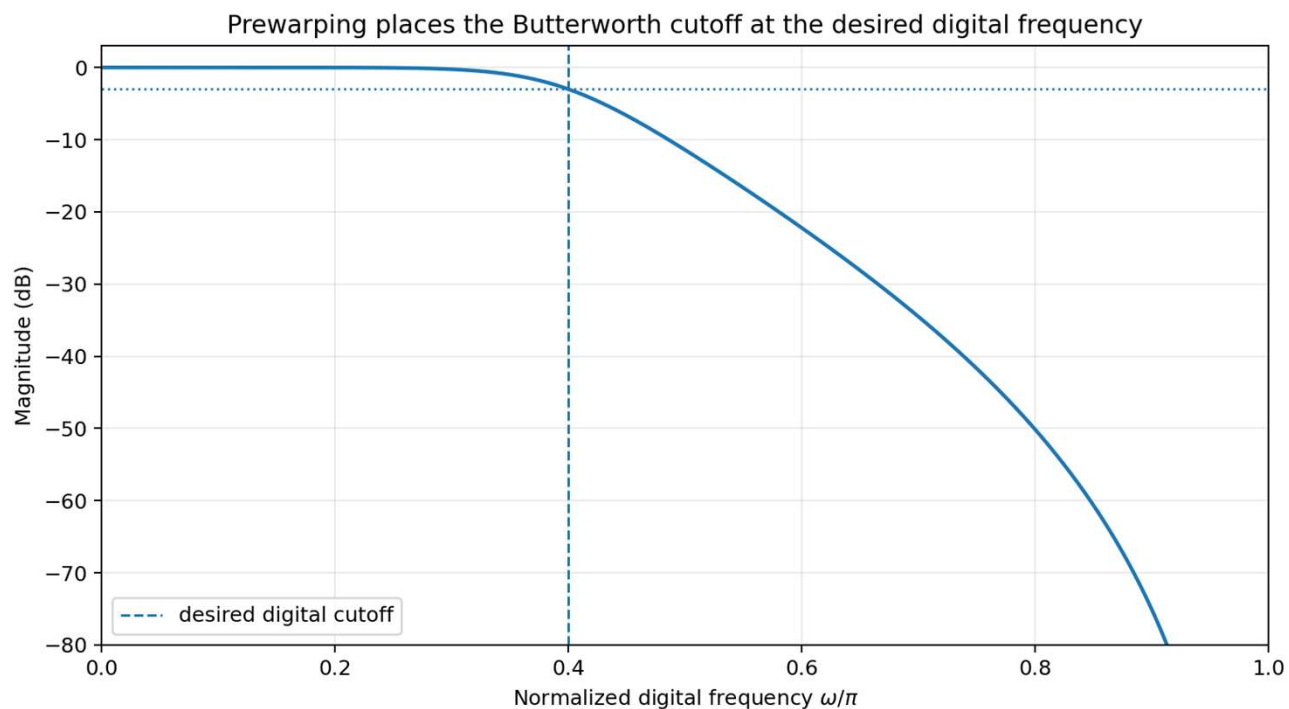
It makes prewarping and the analog prototype visible instead of hiding the mathematics inside one convenience function.

### Implementation practice

For deployment, prefer numerically robust second-order-section forms when order becomes substantial.

## VERIFICATION

# The prewarped design hits the intended digital cutoff



### Target

Desired digital cutoff:  $\omega_c = 0.4\pi$ .

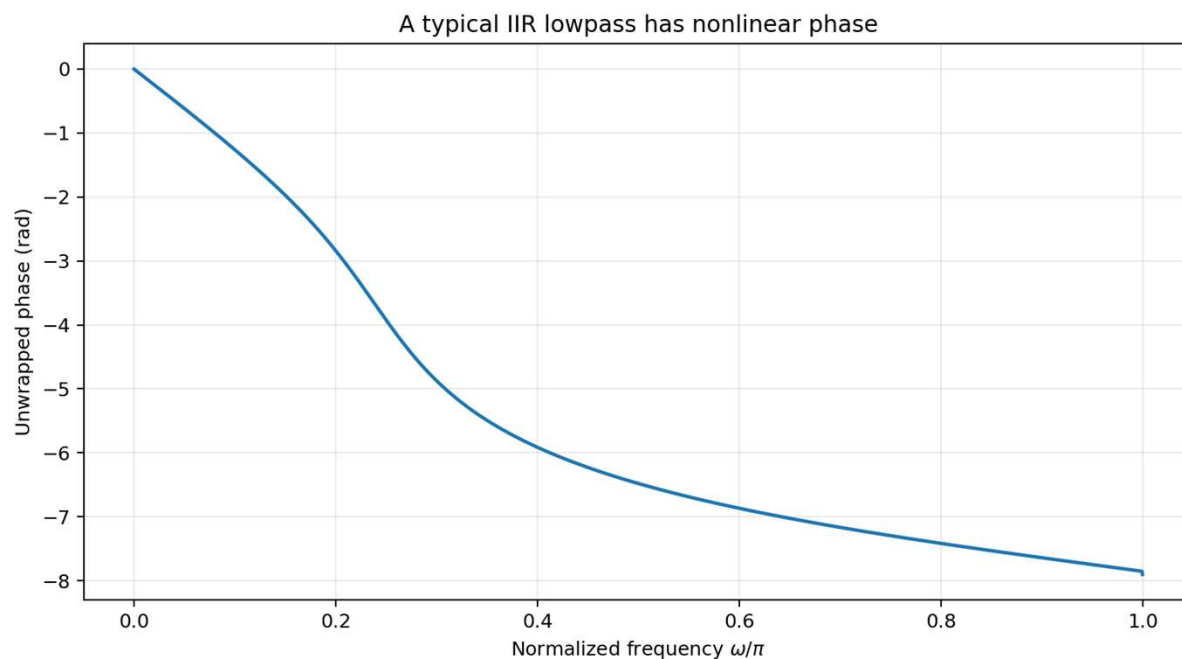
### Measured

The script finds the -3 dB point at  $\approx 0.3999\pi$ .

### Good habit

Always verify the digital response after mapping.

## Efficient magnitude selectivity does not usually preserve waveform shape



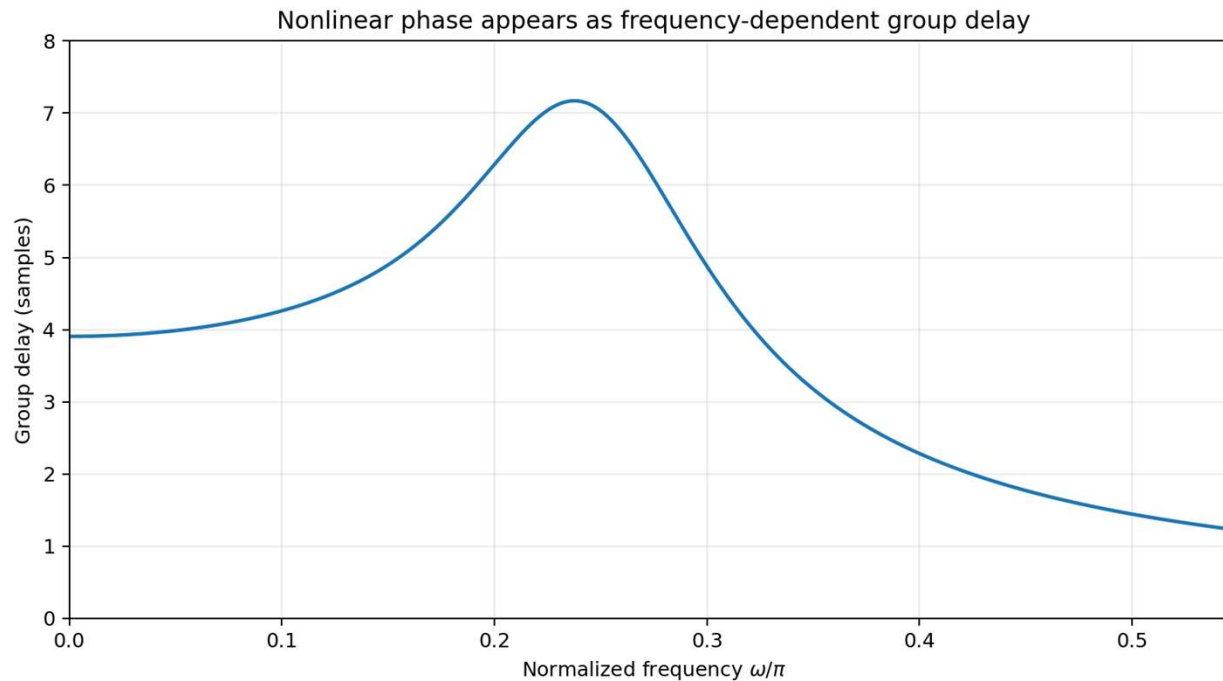
### Nonlinear phase

The phase is not a straight line in  $\omega$ , so group delay varies with frequency.

### ML implication

If timing relationships are part of the feature, causal IIR preprocessing can alter them. That may or may not matter depending on the task.

## Frequency-dependent group delay makes the distortion easier to see



### Contrast with Lecture 7

The symmetric FIR had nearly constant  $\tau_g$ . Here the delay changes across the passband/transition.

### Practical question

Does the downstream model care about absolute timing, relative timing, or only long-term spectral statistics?

## Forward-backward filtering cancels phase

forward:  $H(e^{j\omega})$ ,      backward:  $H(e^{-j\omega})$

$$H_{\text{fb}}(e^{j\omega}) = H(e^{j\omega})H(e^{-j\omega}) = |H(e^{j\omega})|^2$$

### Net phase

For real-coefficient filters, the phases cancel: zero-phase distortion.

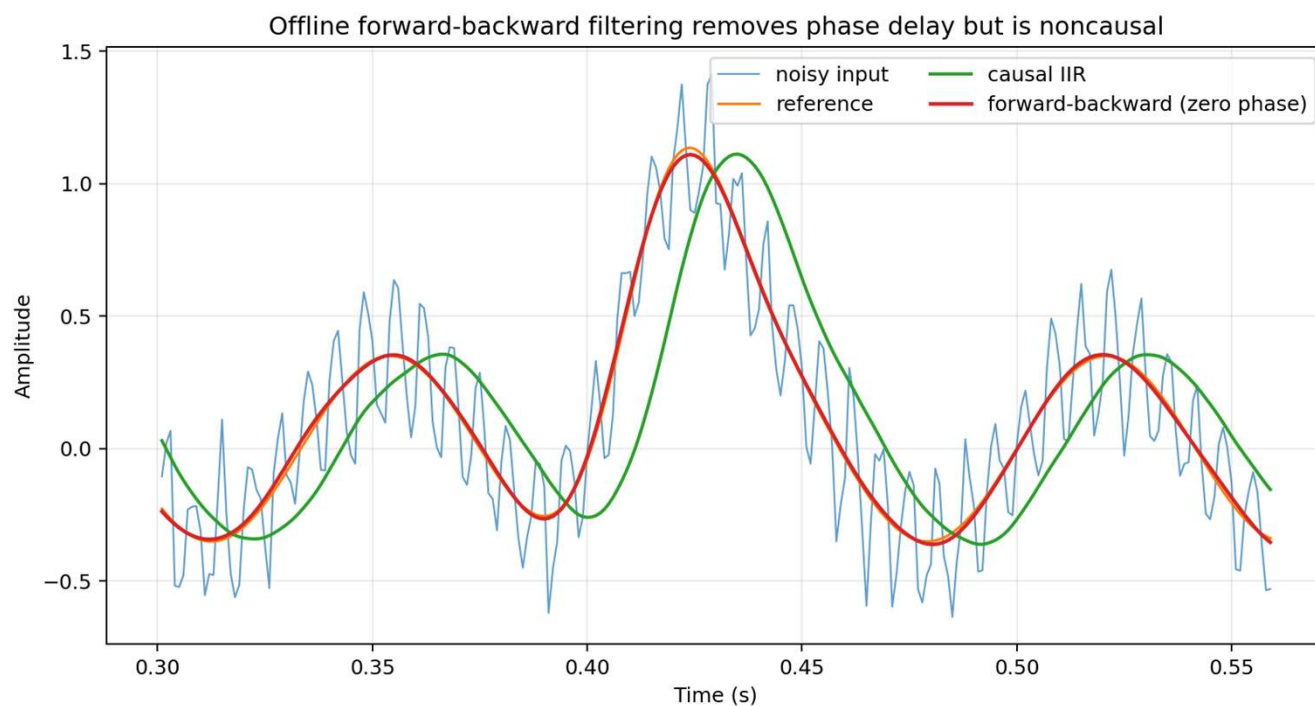
### But magnitude changes

The magnitude is squared, so attenuation in dB is approximately doubled.

### And causality is gone

The backward pass uses future samples relative to a streaming implementation.

## Compare causal IIR delay with forward-backward zero phase



### Causal filter

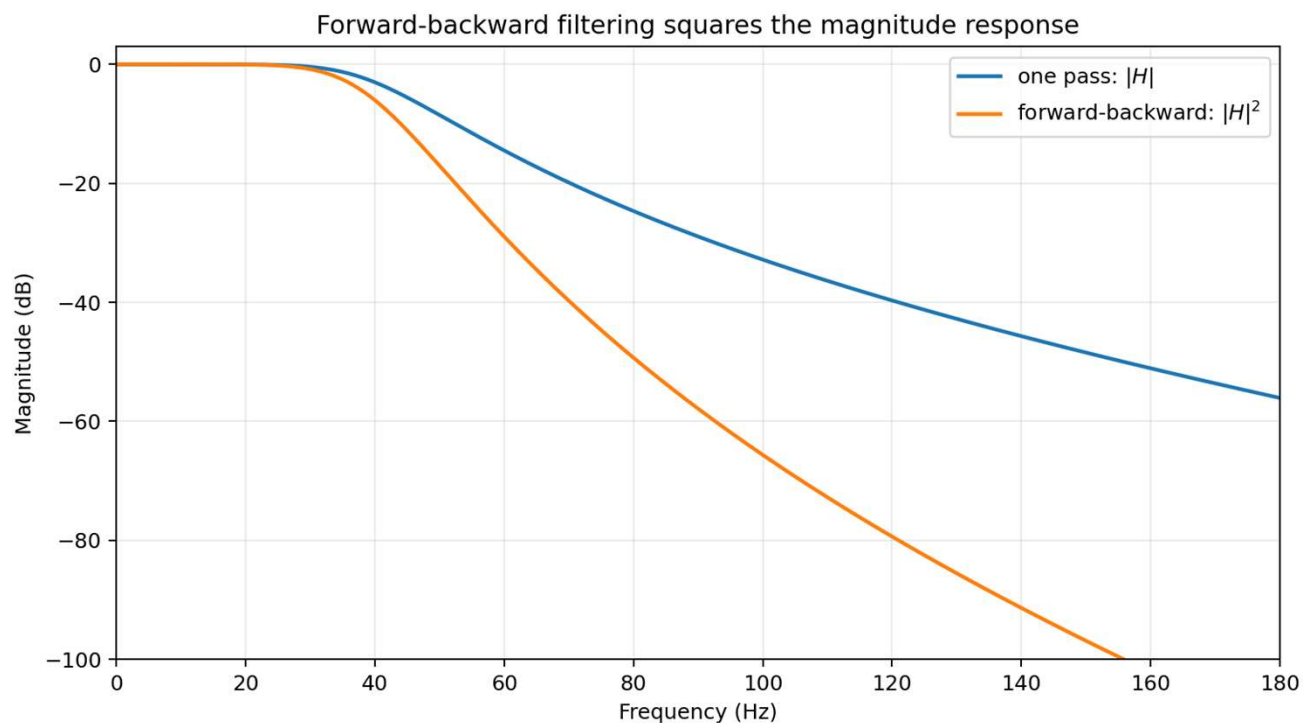
The transient is delayed and reshaped according to the filter phase/group delay.

### Forward-backward

The transient aligns much more closely in time—but the method is offline and noncausal.



## Zero phase does not mean “the same filter twice for free”



### One pass

Magnitude is  $|H|$ .

### Two passes

Magnitude becomes  $|H|^2$ ; order/effective selectivity are increased.

### Design implication

If you need a final  $-3$  dB edge after forward-backward filtering, design with the two-pass response in mind.

## CAVEATS

# Zero-phase preprocessing can quietly change the learning problem

### Noncausal

Future samples are used; not reproducible in a strict real-time stream.

### Boundary transients

Finite-record initial conditions and padding can distort the beginning/end.

### Magnitude squared

The effective response is not the same as one causal pass.

### Train/serve skew

A model trained with future-aware filtering may see a different distribution when deployed causally.

**Good engineering question: “Will the exact same preprocessing be available when the model is actually used?”**

## Where a classical IIR can fit



### Appropriate example

A known high-frequency interference band is irrelevant to the classification task and can be attenuated efficiently before features are computed.

### Question before filtering

Could the “noise” contain class information? Fixed preprocessing is an inductive assumption and should be validated.

## FIR, IIR, or learned front end?

| Choice           | Strength                             | Cost / risk                | Good fit                       |
|------------------|--------------------------------------|----------------------------|--------------------------------|
| Linear-phase FIR | controlled phase + finite support    | more taps / delay          | waveform alignment matters     |
| IIR              | sharp response with few coefficients | nonlinear phase; feedback  | efficient causal preprocessing |
| filtfilt IIR     | zero phase offline                   | noncausal; $ H ^2$ ; edges | offline dataset preparation    |
| Learned Conv1d   | task-adaptive front end              | needs data/validation      | end-to-end models              |

## Lecture 8 synthesis

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### Prototype

Choose where ripple is allowed and how much transition width you need.

### Bilinear map

Stable analog LHP  $\rightarrow$  stable digital unit disk.

### Prewarp

Critical frequencies must be mapped through  $\tan(\omega/2)$ .

### Preprocessing

Phase, causality, edges, and deployment consistency are part of the ML problem.

**Minute paper: Why can a zero-phase offline preprocessing pipeline give misleading confidence about a streaming deployment?**

## HOMEWORK

# Problems 5–8 • IIR, bilinear transform, preprocessing

### 5 • Butterworth

Evaluate a fourth-order Butterworth response at  $\Omega_c$  and  $2\Omega_c$ .

### 6 • Stability map

Map  $s = -2 + j3$  through the bilinear transform for  $T = 0.1$  and verify  $|z| < 1$ .

### 7 • Prewarp

Compute  $\Omega_c$  for desired  $\omega_c = 0.4\pi$ ,  $T = 1$ .

### 8 • filtfilt

Explain phase cancellation,  $|H|^2$ , noncausality, and ML train/serve risk.

Full worked solutions are included in `Lectures7_8_Homework_Quiz_Solutions.md`.

WEEKLY QUIZ

## Questions 1–4 • 10–12 minutes total

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### Q1 • Delay

A symmetric FIR has  $N=31$  taps. What is its group delay?

### Q2 • Window

Hamming vs rectangular: which has lower sidelobes? Which has narrower main lobe?

### Q3 • Gibbs

Explain Gibbs phenomenon in one sentence.

### Q4 • Conv1d

Why flip fixed FIR taps when reproducing textbook convolution?

## Questions 5–8

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### Q5 • Butterworth

Magnitude in dB at  $\Omega = \Omega_c$ ?

### Q6 • Warping

State the  $\Omega$ – $\omega$  bilinear frequency relation.

### Q7 • Prewarp

For  $\omega_c = 0.6\pi$ , should the analog prototype use  $\Omega_c = 0.6\pi/T$  directly? Why?

### Q8 • Zero phase

Main benefit and main deployment limitation of forward-backward filtering?



INSTRUCTOR SOLUTION APPENDIX

# Weekly Quiz + Homework Solutions

Keep these slides after the student-facing material or move them to a separate instructor deck.

## Quiz solutions • Questions 1–4

**A1**

$\tau_g = (31-1)/2 = 15$  samples.

**A2**

Hamming → lower sidelobes;  
rectangular → narrower main lobe.

**A3**

Abrupt truncation produces oscillatory ripple near a frequency discontinuity; increasing  $N$  narrows the ripple region but does not drive peak overshoot to zero.

**A4**

Conv1d uses cross-correlation ordering; textbook convolution contains  $x[n-k]$ , which reverses one sequence.

## Quiz solutions • Questions 5–8

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**A5**

−3.01 dB because  $|H|=1/\sqrt{2}$ .

**A6**

$\Omega=(2/T)\tan(\omega/2)$ .

**A7**

No. Prewarp with  
 $\Omega_c=(2/T)\tan(0.3\pi)$ .

**A8**

Benefit: zero net phase offline.  
Limitation: noncausal/future-aware, so it cannot be reproduced directly in a strict streaming pipeline.

## Homework solutions • Problems 5–6

### 5 • Butterworth

At  $\Omega=\Omega_c$ :  $|H|=1/\sqrt{2} \rightarrow -3.01$  dB. At  $\Omega=2\Omega_c$  with  $N=4$ :  
 $|H|^2=1/(1+2^8)=1/257$ , so  $|H|\approx 0.0624 \rightarrow \approx -24.1$  dB.

### 6 • Pole mapping

For  $T=0.1$  and  $s=-2+j3$ ,  $z=(1+sT/2)/(1-sT/2)\approx 0.785+j0.243$ , so  
 $|z|\approx 0.822 < 1$ .

## Homework solutions • Problems 7–8

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### 7 • Prewarp

For  $\omega_c = 0.4\pi$  and  $T=1$ :

$$\Omega_c = 2 \tan(0.2\pi) \approx 1.4531 \text{ rad/s}$$

### 8 • Forward-backward

Magnitude becomes  $|H|^2$  and phase cancels. It is noncausal, has boundary handling issues, and can create train/deployment mismatch if inference must be causal.